



An empirically-developed framework for Agile transition and adoption: A Grounded Theory approach



Taghi Javdani Gandomani*, Mina Ziaei Nafchi

Department of Computer Engineering, Boroujen Branch, Islamic Azad University, Boroujen, Iran

ARTICLE INFO

Article history:

Received 13 October 2014

Revised 2 June 2015

Accepted 3 June 2015

Available online 10 June 2015

Keywords:

Agile software development

Agile transformation process

Agile adoption

Grounded Theory

Transition framework

ABSTRACT

To date, few Agile transition and adoption frameworks have been proposed in the software industry. However, using them is not easy in practice and primarily requires a huge organizational overhead because of their complex and non-flexible structure. These drawbacks make such frameworks difficult to apply in small and medium-sized companies. We have conducted a large-scale empirical research study using Grounded Theory approach with the participation of 49 Agile experts from 13 different countries. This study inductively developed a substantive Agile transition and adoption framework which appears to be simple and flexible. The main aim of this paper is to present the developed framework. The primary characteristics of this framework, including iterative, gradual, continuous, and value-based are in line with the Agile approach and show promise of being useful in software companies and organizations, regardless of size. This paper also describes how various steps of this framework could help software companies to achieve Agile transformation.

© 2015 Elsevier Inc. All rights reserved.

1. Introduction

Moving from traditional software development methodologies to Agile methods needs much effort and can take a long time (Ganesh and Thangasamy, 2012; Qumer and Henderson-Sellers, 2008). Leaving traditional software methodologies and moving to Agile methods, known as Agile transformation/transition process (ATP), affects almost all aspects of software companies and acts as a revolution in the software development process (Tolfo et al., 2011). In this process, collaboration and involvement of software practitioners such as developers, top and middle managers, and customers is necessary.

A review of previous studies shows that there is no unique model for ATP (Abdelnour-Nocera and Sharp, 2012; Mišić, 2005; Pikkarainen et al., 2012; Qumer et al., 2007; Salo and Abrahamsson, 2007; Sidky, 2007; Srinivasan and Lundqvist, 2010). Also, there is no standard or well-known model or framework that has gained enough acceptances in real environments. The evidence for this claim, besides reviewing the previous studies in the literature, is the increasing number of reports and academic research studies that have been published recently about the challenges and problems of ATP. Indeed, moving to Agile involves a lot of changes in organizational behaviours and, due to the wide extent of the required change, companies are usually faced with many barriers, problems, and challenges

(Gandomani et al., 2013; Nerur et al., 2005; Pikkarainen et al., 2012). Process, people, and management related issues, as well as cultural and technical problems are the main categories of transformation hindrances (Ganesh and Thangasamy, 2012; Nerur et al., 2005; Pikkarainen et al., 2012; Schova, 2010). To ensure a comprehensive ATP, some common questions should be answered. How do software companies and team manage transitioning to Agile? What is an effective and applicable framework for transition? What should be its features and characteristics? What should be its activities? And ultimately, what should be the level of abstraction of the transition framework? Such questions were the main questions that motivate the authors to undertake a large-scale empirical research study.

Finding the best answers to these questions requires substantive data and appropriate research methodology. We found Grounded Theory (GT) to be the most appropriate research methodology to answer these questions. Then, we conducted a GT study to find out the answers of the above questions involving 49 Agile practitioners from 13 different countries. The participants voluntarily participated in this study and had significant experiences in Agile context. Participating in at least on ATP and having experience in Agile methods were two mandatory requirements for attending this research study.

We conducted 49 semi-structured interviews with the participants using open-ended questions. However, since all of them were from different countries, face-to-face interview was not possible. Furthermore, we used supplementary information via memoing in order to find the links and relationships between the emerging concepts. Eventually, we discovered *Agile Transition and adoption*

* Corresponding author. Tel.: +98 9131065201.

E-mail address: tjavadani@yahoo.com, t_javadani@azad.ac.ir (T. Javdani Gandomani).

framework and its related properties by meticulously following GT procedure.

While most of the attendees confessed that they had never used a well-defined or pre-defined transition framework, some of them claimed that they had used their own experience-based frameworks. At the same time, most of them addressed the lack of a well-defined framework as a significant barrier in ATP. Furthermore, the participants addressed the required activities of transition framework and explained their problems and challenges during ATP in real environments.

This paper presents the following contributions: (1) a description of a substantive framework for Agile transition and adoption, its structural characteristics, and its required activities, (2) application of a GT approach to develop a substantive theory, (3) theoretical and practical implications of the results.

The rest of this article is organized as follows: The next section provides a short background on Agile methods and Agile transformation process challenges and issues. Section 3 explains the related works. In Section 4, we describe the employed research methodology in detail. Section 5 presents the results of the study. Subsequently, in Section 6, we provide a detailed discussion including comparison of our findings with related works, and implications of our findings for theory and practice. Section 7 presents the evaluation process of the results. Sections 8 and 9 respectively identify the limitations of the study and future works. Finally, the last section concludes the paper.

2. Background

Agile methods are introduced as reactions to traditional methods in software development, primarily to overcome the inherent challenges of traditional methods (Cohen et al., 2004). Compared to traditional methods, Agile methods offer different values including early and frequent delivery, light-weight documentation, embracing change in requirement, customer satisfaction, high quality development, etc. (Cohen et al., 2004). Although some of these new methods had been developed in the late 90s, they were formally introduced to the software industry by gathering under the Agile umbrella when the Agile manifesto was created in 2001 (Beck et al., 2001). Extreme Programming (XP) (Beck and Andres, 2004), Scrum (Cohn, 2009), lean software development (Poppendieck and Poppendieck, 2009), Feature-driven development (FDD) (Highsmith, 2002), Crystal methods (Highsmith, 2002), Dynamic systems development method (DSDM) (Highsmith, 2002), Test-Driven Development (TDD) (Beck, 2003), are some of the well-known Agile methods. These methods are different in nature, goals, and activities. However, they are all focused on similar values.

ATP encompasses all practices and activities necessary to change software development approach from traditional to Agile. The scope of these activities is not limited to technical practices only and generally covers social behaviours and human aspects too (Cockburn and Highsmith, 2001; Iivari and Iivari, 2011; Tolfo et al., 2011).

So far, many research studies have been conducted to investigate ATP and its related challenges and issues (Conboy et al., 2011; Hajjdiab and Taleb, 2011; Nerur et al., 2005; Srinivasan and Lundqvist, 2010). However, only a few strong academic studies can be seen between them. Most of the related publications primarily have described the introduction of Agile methods to software companies and explained the main challenges and barriers they have been faced with during ATP.

Only a few studies have focused on proposing a transition framework to facilitate ATP and the lack of a comprehensive and effective framework to support ATP has been highlighted in many studies (Chan and Thong, 2009; Esfahani, 2012; Hajjdiab and Taleb, 2011; Rohunen et al., 2010; Srinivasan and Lundqvist, 2010; Vijayasarathy and Turk, 2012; Wang et al., 2012). At the same time, there are sev-

eral criticisms against the proposed frameworks which most of them are concerned mainly with their complexity or usability as well as the inconsistency of their nature with Agile approach (Esfahani, 2012; Soundararajan and Arthur, 2011; Soundararajan et al., 2012). The raised criticisms have been elaborated in Section 3. It should be noted that though Agile methods were introduced in the early 2000s, their increasing prevalence has been seen in recent years (VersionOne, 2014; West et al., 2010). This denotes that most of the previous models and frameworks have not come from empirical or large-scale research studies, and most often they have not been industry supported (Misra et al., 2009; Vijayasarathy and Turk, 2012). Therefore, adopting an empirical approach to study ATP would be beneficial.

Focusing on the above-mentioned gap, we tried to focus on this problem through a large-scale research study. Prior knowledge of researchers and its influence during data analysis and theory generation is considered as a risk factor in qualitative research studies and delayed literature review is recommended to decrease this adverse effect (Giles et al., 2013). The employed research methodology advised us against undertaking an up-front extensive literature review, mainly to prevent biases in the questions or demographics of the participants (Glaser, 1992). However, as Glaser (2005) explained, applying GT is not an excuse to ignore conducting literature review. Literature review is considered as another source of data when applying GT (Glaser, 1992). Therefore, we started this study with a minor literature review and carried out an extensive literature review when emergent categories and their properties were sufficiently developed and grounded in data. This mainly helps GT researchers to tie the results into existing literature (Hoda et al., 2011a). The next section of this paper provides a literature review for the benefit of readers and in keeping with the order of the major literature review, the research results are followed by a discussion of existing literature in Section 5.

3. Related works

So far, a small number of transition frameworks have been proposed for Agile transition and adoption. Although there are several studies providing recommendations and lessons learned to facilitate ATP, only a few structured Agile transition and adoption frameworks can be considered (Esfahani, 2012; Rohunen et al., 2010; Vijayasarathy and Turk, 2012).

Boehm and Turner (2003) proposed a simple framework that directs organizations to choose either Agile methods or disciplined methods or a mix of both. They had defined five critical decision factors: size, criticality, culture, personnel, and dynamism. They also proposed a stepwise approach for assessing organization and project that assists organizations to decide about rebalancing their agility and discipline by rating those critical factors. A risk-based method was the core of their proposed framework. Although, this framework did not explain how to adopt methodologies, acceptance of Agile methods by advocates of disciplined methods and considering human factors in method selection were interesting at that time.

Sidky et al. (2007) proposed a multi-stage (like CMMI model) transition framework for Agile adoption. They also proposed an Agile measurement index for evaluating and indicating the maximum potential Agility degree of an organization. The framework was defined based on the proposed index and its first step was identifying discontinuing factors, which were defined as indicators, the existence of which showed the unreadiness of organizations for Agile adoption. In the next step, both organization and project should be assessed to evaluate their capability for Agile adoption. Finally, based on organization and project capability level, they have suggested the most appropriate and suitable set of Agile practices for adaptation. Despite the novelty of this framework in proposing a set of Agile practices with respect to the potential Agility of an organization, the framework is subject to serious criticisms in the empirical world.

The framework defined more than 300 indicators for Agility assessment and using this huge number of indicators not only is time consuming, but also needs a dedicated expert to handle the measurement process. Furthermore, applying such a comprehensive, complex, highly-disciplined, and detailed framework can lead to significant overhead and is completely inconsistent with the flexibility promised by Agile approach. One of the authors of the framework, 4 years later, criticized it and explained that forcing companies to consider the Agility levels and specific practices for each level compromises the flexibility offered by Agile methodologies (Soundararajan and Arthur, 2011). All the aforementioned consequences make it difficult to use in practice, especially in small and medium-sized companies. Ultimately, there is no guarantee that the proposed practices will be the most suitable set for an organization (Esfahani, 2012).

Qumer and Henderson-Sellers (2008) proposed an Agile adoption framework, which was backed by an Agility measurement technique. The framework, called Agile Adoption and Improvement Model (AAIM), defines six levels of Agile adoption that are respectively, Agile Infancy, Agile Initial, Agile Realization, Agile Value, Agile Smart, and Agile Progress. At each level, AAIM seeks to incorporate certain Agile principles (e.g. Speed, Flexibility, People orientation, Leanness, etc.) into the organization. This framework uses the 4-DAT Agility measurement technique, which provides specific metrics for quantitatively assessing the Agility level of a software process. Unlike Sidky et al.'s framework, AAIM does not explicitly propose Agile practices to an organization; rather it focuses on localization of Agile principles. However, the reliance of this framework on a quantitative measurement scale makes it vulnerable to correctness of accumulative numbers that are supposed to represent the Agility degree of a process. Like previous models, applying this model in small and medium-sized organizations is not easy or affordable. Furthermore, forcing companies to consider various Agility levels reduces flexibility and is inconsistent with the Agile approach (Soundararajan and Arthur, 2011).

Sureshchandra and Shrinivasavadhani (2008) proposed a framework for Agile adoption in distributed development. The framework is supported by a tool, which is based on the initial framework proposed by Boehm and Turner (2003), for evaluating the degree of Agility and formality that is needed for a project. The framework is composed of four stages: Evaluation, to determine the degree of distribution of the project; Inception, to form the distributed teams, and the way Agile practice can be incorporated into the development infrastructure; Transition, to enact the Agile practices into the development activities; and Steady State, to provide baselines for smoothly adding further distributed teams into the project. While the researchers have focused on distributed teams and dealing with their special issues, they have paid less attention to providing directions for adapting Agile practices in the team level. Indeed, they have mainly tried to apply Scrum in a distributed environment rather than propose a method-independent transition framework.

A similar concept to the Agile adoption is studied under the Agile adaptation framework. Cao et al. (2009) proposed a framework for adapting Agile development methodologies, which is theoretically based on Adaptive Structuration Theory (AST) (Jones and Karsten, 2008) and four case studies. The framework investigates the adaptation of Agile methods with respect to the project and organization factors. The main issue is that the main focus applies AST in the context of software development and as the authors confessed, application of AST needs to be validated (Cao et al., 2009). Indeed, the proposed framework is an extension of AST. At the same time, it seems that the selected case studies were not mature enough in Agile development and Agile practices, since they (three companies) had been using only some of the XP practices. However, the framework was a step forward as it was more Agile than the previous ones.

We tried to focus on ATP to inductively develop a transition framework grounded in real practices. Although GT does not involve formu-

lating research hypothesis up-front and based on an extensive literature review, application of literature review is not forbidden. However, Glaser (1978) strictly warned GT researchers about the adverse effects of conducting an extensive literature review in the same area of research during early stages of GT. Following Glaser's recommendation, we started this research study by conducting only a minor literature review in the beginning. We primarily wanted to understand general concepts, terminology, and basic facts in order to be able to carry on direct conversations with the interviewees.

4. Research methodology

This empirical study was carried out by employing Pure GT, known as Glaser version. This research methodology was developed by Glaser and Strauss (1967). It provides a systematic approach to inductively build a theory on the basis of data analysis (Glaser, 1978, 2005). It uses a "systematic set of procedures to develop an inductively-derived grounded theory¹ about a phenomenon" (Strauss and Corbin, 1990). GT is a suitable method for qualitative researchers to answer questions like "what is going on in an area?", by generating formal or substantive theory (Corbin and Strauss, 2008). Although this method has been used in social studies, it is a useful method for a wide range of topics in software engineering too; especially those related to people and human aspects (Hoda et al., 2010). Fig. 1 shows the steps that have been taken in this study.

4.1. Why Grounded Theory?

This methodology was chosen for conducting this study for multiple reasons as follows:

- (1) The main focus of Agile approach is people rather than processes. In fact, all Agile practices and values rely on people. On the other hand, GT is a research method that assists researchers to study people interactions and behaviours (Hoda et al., 2010; Parry, 1998). Therefore, using GT to study a people-oriented process would be suitable choice.
- (2) GT is very helpful in studying a phenomenon in detail and in substantive areas. Also, it is very helpful to study a relatively new area or when researchers try to get a "fresh perspective on a well-known area" (Stern, 1994). In a few related works only a few theoretical frameworks for Agile transition have been proposed but none of them have support from real environments and different perspectives.
- (3) While extending or verifying existing theories is the main purpose of almost all research methodologies, the main purpose of GT is theory generation; which is an exciting and interesting prospect.
- (4) Fortunately, several valuable GT studies had been conducted on Agile software development in recent years and all of them had shown that GT produces reliable results in Agile software development (Coleman and O'Connor, 2007; Ghanam et al., 2012; Hoda et al., 2011a, 2011b; Martin et al., 2009). It seems that this methodology has been frequently used successfully, and still continues to gain acceptance and popularity, as a qualitative research methodology to study Agile context (Woodman and Moteleb, 2009).

GT works well especially when there is no clear hypothesis or research problem up-front. In this case, researchers try to discover the main concerns of participants in a specific area under study (Parry, 1998). Following the GT guidelines, we started this study in a wider area, Agile transition strategy, rather than focusing on a particular research problem (Coleman and O'Connor, 2007). Also, we continually employed constant comparison technique, as explained later,

¹ "Grounded Theory" refers to the research methodology, while "grounded theory" refers to the outcome of the research process which is an emerged substantive theory.

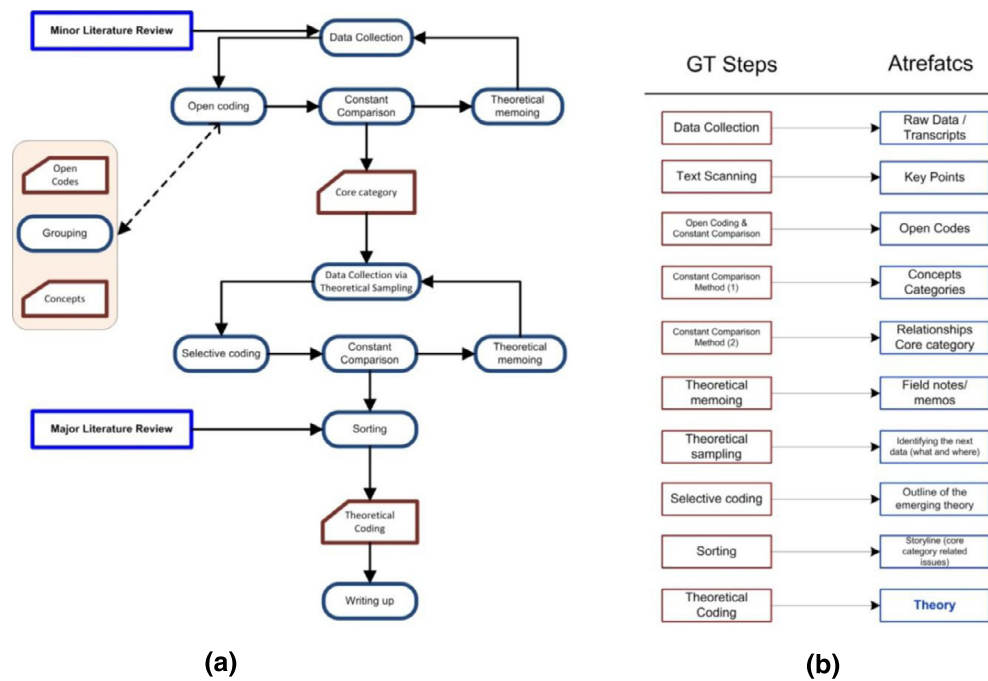


Fig. 1. (a) GT steps have been taken in this study, adapted from (Hoda et al., 2011a), (b) Artefacts of a GT study.

to achieve a high level abstraction in findings. Eventually, grounded theory emerged and developed by finding the relationship between discovered main hypotheses (Glaser, 1992).

It should be noted that because of a variety of reasons, we did not employ other qualitative research methodologies. For example, we had no clear question or hypothesis up-front, so, Root Cause Analysis (RCA) and Survey Research were not applicable for this study (Creswell, 2013; Latino et al., 2013). Also, since Agile transition is considered a socio-technical process and particular organizational or cultural behaviours of one or more case studies may not be generalized to other groups, Case Study was not an appropriate choice to be employed in this study (Yin, 2013). Moreover, a researcher using Ethnography often becomes a member of the community for the duration of the observations (Hammersley and Atkinson, 2007). This requirement also was not applicable for this study.

4.2. Data collection

Typically, a GT starts with data collection and uses it as an ongoing activity to reach data saturation—the point at which no new concepts or information is being obtained (Glaser, 1998). After finding some concepts representing the main concerns of participants, next data can be collected via *theoretical sampling*, which means that the researcher can decide what data should be collected in the next steps (Glaser, 1978). We conducted all the interviews based on this approach.

4.2.1. Recruiting participants

We published an invitation for participation in the study in the most reputed and populated on-line Agile communities and groups on Yahoo and Google, and some local networks. In this invitation, we called for those people who had experience in both traditional and Agile software development and had participated in at least one Agile transformation project. Although this requirement was compulsory, determining qualification of the participants was not possible before attending to the research study. In fact, we trusted those who claimed to be qualified for this study, mainly because there was no initial way to verify their claims.

Ascertaining authenticity of the participants' viewpoints has been a topic of debate in qualitative research studies (James and Bushe, 2006). However, the iterative discussions on the emerging concepts of the interviews were the only way that allowed us to interrogate the authenticity of the participant's information (James and Bushe, 2006). By using this strategy, later we found that almost all of the volunteers had a good background in the context under study. We also tried to choose interviewees from different countries and in different roles in Agile software development organizations, primarily due to the advisability of covering cultural related issues and considering different perspectives in the study.

Forty-nine (49) participants from 49 different companies in 13 different countries participated in this study. Table 1 shows the brief information about the participants. For simplicity, in this article they are referred to by their numbers and positions only. Geographical distribution of the participants relatively supports the previous study on Agile distribution (Dingsøyr et al., 2012). They were using mostly a combination of Scrum, XP, and Lean as Agile methodologies that nowadays are more popular than the others (VersionOne, 2013). Size of the companies varied from 10+ to 5000+, with most of them having more than one branch. Transformation process in about half of them was ongoing and this provided us with the opportunity to collect fresh data. The participants' roles varied and included senior manager, project manager, Scrum master, developer, coach, consultant, etc. However, most of them were coaches and this provided a great opportunity for this study to use their previous valuable experiences in several companies and organizations. Data collection through theoretical sampling besides helping in adopting questions with emergent concerns, helped us to choose more eligible participants for the next interviews (Glaser, 1978).

4.2.2. Interviews

We collected the data by conducting semi-structured interviews with the participants using open-ended questions. With one exception, the interviews were conducted in English. Each interview took about an hour and usually started with some general questions about the interviewee's experiences and his or her background on ATP, then focused on challenges faced during transformation, their used

Table 1

Demography of the participants (Agile positions: Agile Coach (AC), Scrum Master (SM), Project Manager (PM), Developer (DEV), Consultant (CON), Head of Development Dept. (HDD), Manager (MGR), Quality Assurance (QA)).

Participant number	SD exp. (years)	Agile exp. (years)	Agile position	Agile methods	Country	Projects domain	Branches	Emp. No.	Transformation period (months)
P1	14	8	HDD	XP, Scrum, Kanban	Finland	Mobile software	1	10+	12+, ongoing
P2	25	15	AC	Scrum, Kanban	USA	All domains	4	65+	12+
P3	7	7	PM	XP, Scrum, Kanban	USA	Web applications	3	500+	6+, ongoing
P4	10	2	PM	XP, Scrum, Kanban	Bulgaria	Banking	1	200+	6+, ongoing
P5	10	2	PM	Scrum, Kanban	Iran	Financing	7	150+	12+, ongoing
P6	11	8	CON	Scrum, Kanban, FDD	Australia	All domains	7	1000+	12–15
P7	6	2	DEV	Scrum	Greek	Office auto.	1	20+	12+, ongoing
P8	10	5	PM	Scrum, Kanban	Germany	Web applications	5	50+	8+, ongoing
P9	20	10	HDD	Scrum	Spain	Custom development	3	200+	24
P10	20	3	SM	Scrum, Kanban	Spain	Transportation	4	200+	24+, ongoing
P11	10	4	AC, SM	XP, Scrum, Kanban	India	Publication	1	50+	+6, ongoing
P12	16	2	HDD	Scrum, Kanban	USA	Life and insurance	16	1600+	6+, ongoing
P13	14	6	AC, CON	Scrum, Kanban	Finland	Consultancy services	1	20+	3–30
P14	15	3	MGR	Scrum, Kanban	Iran	HRMS, CMMS	1	50+	12
P15	10	2	CON	Scrum	Indonesia	MIS	1	200+	3+, ongoing
P16	21	10	PM	Scrum, Kanban	USA	SCM	1	65+	12
P17	19	5	PM	Scrum, Kanban	Sweden	Financing	2	50+	24+, ongoing
P18	8	2	DEV	Scrum	Sweden	HRMS, MIS	1	40+	24
P19	13	6	PM	Scrum	India	Banking	3	200+	18 in USA, 24 in India
P20	11	3	HDD, PM	Scrum, Kanban	USA	Insurance, banking	16	1200+	6+, ongoing
P21	16	7	SM	XP, Scrum	USA	Financing, HR	3	250	18
P22	11	5	AC	Scrum, Kanban	France	Banking	100+	2000+	12+, ongoing
P23	16	8	AC	XP, Scrum, Kanban	USA	All domains	7	200+	6–24
P24	15	7	SM	Scrum, XP	USA	All domain	3	100+	12
P25	8	4	DEV	Scrum, XP	USA	HRMS	1	40+	6+, ongoing
P26	13	6	AC	Scrum, XP	India	Medical systems	3	300+	15+
P27	14	5	SM	Scrum, Kanban	USA	Mobile applications	2	50+	12+
P28	15	6	AC	Scrum, Kanban	Germany	Advisory/consulting	2	50+	15+
P29	10	1	PM	Scrum	Norway	Web agency	2	40+	12+
P30	35	1	DEV	Scrum	USA	Finance, banking	3	100+	6+, ongoing
P31	17	4	QA, PM	Scrum	USA	All domains	3	50+	12
P32	25	2	AC	Scrum, Scrumban	USA	All domains	2	200+	12, ongoing
P33	41	3	MGR	Scrum, Kanban	USA	All domains	17	2500+	15+, ongoing
P34	7	2	QA	Scrum, Kanban	USA	Retail systems	3	100+	9+, ongoing
P35	20	8	HDD	Scrum, XP	USA	Life and Annuity	50+	2000+	15
P36	17	7	MGR	Scrum, Scrumban	USA	Investment services	21	5000+	18
P37	13	1	HDD	Scrum	USA	Finance, banking	3	200+	6, ongoing
P38	7	2	PM	Scrum	USA	Web applications	1	50+	12+, ongoing
P39	35	10	AC	Scrum, XP, Kanban	USA	Consultancy services	1	20+	15
P40	30	15	AC	Scrum, Kanban	USA	Consultancy services	1	50+	12
P41	11	3	PM	Scrum, Kanban	India	Mobile applications	2	150+	12+, ongoing
P42	14	4	DEV	Scrum	Spain	Banking, HR	14	2000+	15+, ongoing
P43	18	6	AC	Scrum, XP	USA	Banking	2	150	6+, ongoing
P44	17	5	SM	Scrum, Kanban	Norway	HRMS	2	60+	9+
P45	11	6	PM	Scrum	USA	Mobile applications	4	350+	12+
P46	14	3	PM	Scrum, XP	USA	All domains	2	100+	12+, ongoing
P47	11	2	DEV	Scrum	USA	Mobile application	1	50	3, ongoing
P48	22	8	AC	Scrum, Kanban	Spain	Banking	1	80	15
P49	17	6	SM	Scrum	USA	Financial systems	2	130+	6+, ongoing

solutions and suggested strategies for handling issues. The next set of questions was about characteristics of ideal or effective Agile transformation process and role of the effective factors in it (e.g. framework, method selection, pilot project, customer, management, people, tools and technology, etc.).

With the interviewees' consent, all interviews were voice recorded and transcribed for further analysis. Later, we used the transcribed data for analysing and reviewing in several iterations, as GT suggests, for achieving the goal of constant comparison, as explained later (Glaser, 1998). Also, based on GT rules, we never asked any direct questions about very specific items (Glaser, 1992). Using the iterative method for reviewing the transcribed data, we placed emphasis on some specific questions in the next interviews.

4.3. Data analysis

Data analysis procedure, called *data coding* in GT, started when some data were collected and continued up to emerging theory

(Glaser, 1998). Data coding involved *substantive* and *theoretical* coding as two different approaches. Substantive codes are the emergent categories and properties that conceptually describe the phenomenon under study. On the contrary, the theoretical codes are the emergent abstractions that model the integration of substantive codes as an interrelated set of hypothesis for resolving the main concern (Glaser, 1978, 2005). It means that by such codes, GT researchers are looking to find relationships between concepts and categories and modelling hypothesis (Glaser, 2005). Applying both approaches, we employed a multi-level coding process that is explained in the rest of this section.

4.3.1. Tool support

Computer-Assisted Qualitative Data Analysis Software (CAQDAS) has been seen as aiding qualitative researchers in facilitating data collection, handling, and analysing, and ruling out or reducing human error when working with large amount of data (Kelle, 1995). CAQDAS has been used to facilitate GT since the early 2000s (Bringer et al., 2004; di Gregorio, 2003) with the first version of QSR-NVIVO

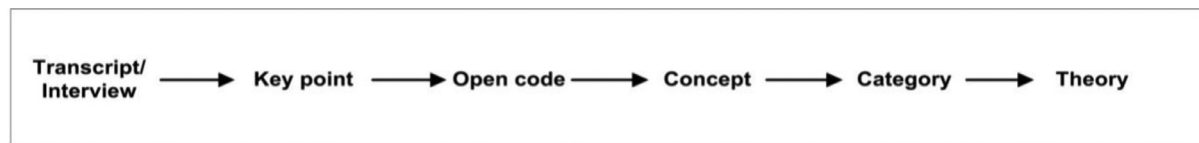


Fig. 2. Data analysis artefacts in a Grounded Theory research.

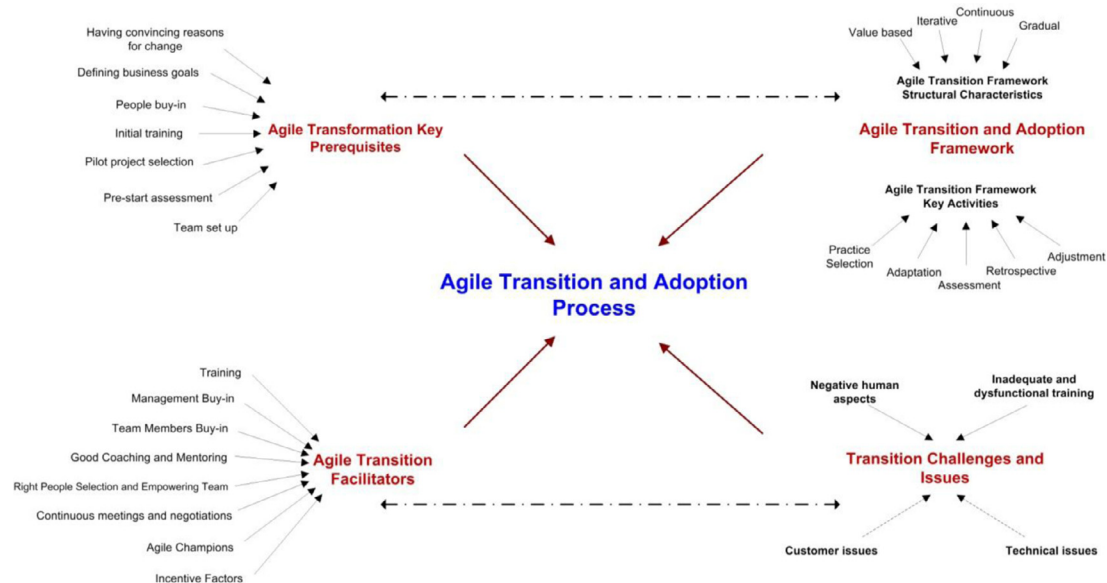


Fig. 3. Emergence of the grounded theory of Agile transition and adoption from underlying concepts.

package. Later, the advanced versions of this software helped many GT researchers (Bringer et al., 2006; Hutchison et al., 2010). In this study, we used NVIVO software to facilitate data analysis and manage the GT required steps. Fortunately, as the previous researchers addressed (Bringer et al., 2006; Hutchison et al., 2010), we found it very helpful to handle a huge amount of data.

4.3.2. Open coding

The first step of data analysis was *open coding* in which each transcript was analysed sentence-by-sentence or line-by-line for conceptual understanding of the context under study (Glaser, 1978). By employing this method, we were looking for *key points*. Immediately after finding a key point, a code was assigned to that key point (Glaser, 1978). Afterwards, the emerged code was compared with the previous open codes in the same and previous transcripts. This process is known as *constant comparison* process and the heart of GT (Glaser and Strauss, 1967). Using NVIVO and its facilities for constant comparison led to a higher accuracy in data analysis (Hutchison et al., 2010). Constant comparison technique helped us to achieve abstraction in a higher level by finding *concepts* (Glaser, 1978). Applying iterative constant comparison on the emerging concepts helped to find *categories* (Glaser, 1978). Each category is a higher level of abstraction, which encompasses a group of concepts that seem to relate to the same phenomenon. The final theory, as explained later, would be based on some or all of the emergent categories. Fig. 2 shows the multi-level data analysis in a GT study.

4.3.3. Core category

The end of open coding is marked by the emergence of a *core category* (Glaser, 1992). The core category “accounts for a large portion of the variation in a pattern of behaviour” and is considered the main problem or concern of the participants (Glaser, 1978). A challenge for GT researchers, however, is that discovering a core category can be time consuming and tedious.

Several criteria have to be considered to identify a core category. Core category must be central; it must be related to several other categories and their properties; it must re-occur frequently in the data; it must relate meaningfully and easily with other categories; and it must account for most variations in data (Glaser, 1978). Using constant comparison technique on the emergent categories and discovering their relationships helps GT researchers to find core categories. In this study, using NVIVO helped us to identify the relationships between codes and setting them in separate groups (Glaser, 1992). Once a category emerges, GT researchers need to identify its properties and related categories. They also need to show how categories vary dimensionally along properties (Corbin and Strauss, 2008).

In this research, in almost half of the research time, the category *inadequate and dysfunctional training* seemed to be the core category, because it was the most common concerns of all the research participants, as we have explained it in (Gandomani et al., 2015). Focusing on the core category criteria is helpful for dealing with this challenge. In this case, *inadequate and dysfunctional training* did not meet some of the above criteria. For instance, although it was a critical concern of the participants, it was not central. Also, by growing the amount of data, it did not account for the most variations in data. Continuing data analysis showed that the core category of this study was *Agile transition and adoption process* and *inadequate and dysfunctional training* was a serious challenge faced by software practitioners while resolving their main concern. Indeed iteratively conducting constant comparison pushed the discovery of ‘*Agile Transition and Adoption Process*’ as the core category of this study whereby ‘*Agile Transition and Adoption Framework*’ was one of its related categories. Fig. 3 depicts the emerged core category of this study. Once we established the core category, open coding was stopped. Afterwards, the coding process continued only for the core category and its closely related categories (Glaser, 1978). This process is known as *selective coding* in GT.

The emerged grounded theory of this study comprises four major parts, including “Agile Transition and Adoption Process”, “Transition key prerequisites”, “Transition challenges”, and “Transition facilitator”. This paper is aimed at describing the developed framework only rather than the emerged core category, i.e. *Agile Transition and Adoption Process*. The core category will be described later on in another paper.

4.3.4. Theoretical memoing

After each interview, on the basis of the emphasised ideas and concepts, more detailed data were collected and formed in one or more memos related to that interview. This process is known as *theoretical memoing* (Glaser, 1998). Each memo is simply a record of feeling, idea, thought, and insight in relation to an emerging concept. Memoing is not optional and is the most significant factor in ensuring quality in GT (Birks and Mills, 2011). This technique helped us to enrich relationships between codes. In fact, when we had an idea about the emerging open codes, concepts, and categories and their relationships, we wrote our idea as a memo and if necessary, we collected more data for the memos. It should be noted that we did not mix memos with the collected data. Also, we used NVIVO to handle memoing in this study.

4.3.5. Sorting

Sorting is defined as the process of formulating the emergent theory and its related categories and links. This process was started when almost all codes had been saturated, and data collection was nearly finished. This step was very important, since it was the last step of the analysing process and helped us to explain each topic in detail (Corbin and Strauss, 2008). Using this procedure assisted in the formation of structure and outline of the Agile transformation strategy.

4.3.6. Theoretical coding

Theoretical coding or theory building was the last step of this study and was used to look for connections and relationships between the core category and other emerged categories, and form the hypotheses that together explain the theory (Glaser, 1992; Glaser and Strauss, 1967). In this stage, the relationships between *Agile Transition and Adoption Process* as the core category and other categories such as *inadequate and dysfunctional training*, *Agile transition and adoption framework*, *transition facilitators*, and *transition challenges* were formulated. As explained earlier, Fig. 3 depicts the emerged core category and its related categories; each includes its own properties. In this paper, as explained earlier, only *Agile Transition and Adoption Framework* is described.

Glaser suggests several structures of theories, called ‘*theoretical coding family*’ (Glaser, 1978). One of the most popular codes in this family is about the process concept. He defined this family as *Process Family* at first (Glaser, 1978) and later he elaborated it and called it *Temporal Family* (Glaser, 2005). We used this coding family to formulate the emerging transition framework. This coding family demonstrates a category in the form of stages, phases, steps, timelines, conditions, actions, temporal ordering of work, and so forth.

In the next section, the emerged framework will be explained in the form of the Process/Temporal coding style. Fig. 4 depicts how category of *Agile Transition and Adoption Framework* has been illustrated in the form of Glaser’s Temporal/Process Family (Glaser, 1978, 2005).

5. Results

In this section, the emergent transition framework is explained by illustrating its related emerging category. As Fig. 4 depicts, the emergent category describing *Agile transition and adoption framework* comprises two distinct parts; structural characteristics and key activities. The rest of this section describes these items in detail. Due

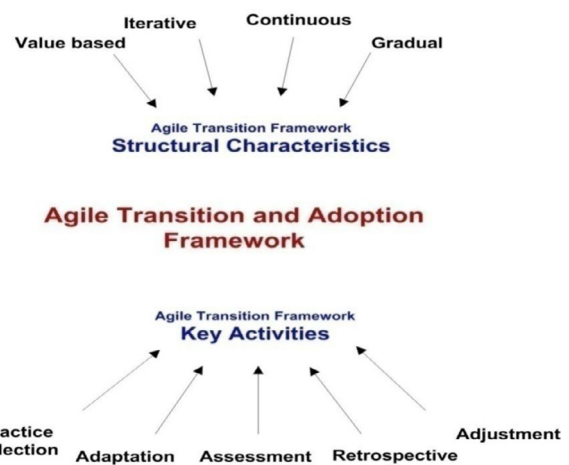


Fig. 4. Emergence of the Agile transition and adoption framework characteristics and activities from underlying concepts.

to the limited space, only the most important derived quotations that depict some parts of the theory are presented.

5.1. Structural characteristics of the transition framework

Data analysis revealed the most important characteristics that an applicable transition framework must have. The participants believed that considering these characteristics as essential features of the transition framework, makes it less costly and time demanding, and more effective.

5.1.1. Value based

Some of the more experienced participants emphasised that *Agile* transition framework should consider ‘*business values*’ in its core. Considering business values and focusing on achieving more business values among the transition, are the most important factors which lead to successful transition. *Agile* transition framework needs to help software companies to achieve, as much as possible, business values.

“It’s very important to have clear goals why you are adopting Agile. The main mistake is starting a process because of process and use a practice because of practice. Values should shade all of the transformation tasks. Framework should be value oriented rather than practice oriented.” P40, Agile Coach.

Following *Agile* values and putting emphasis on them were stressed by almost all of the participants. They explained that visibility of values and defining appropriate metrics for measuring expected values are two significant factors in designing an effective framework.

“We intend to use Agile methods only because they can create new values, thus, the values should be seen and felt in the proposed framework.” P16, Project Manager.

At the same time, some of them warned about lack of business benefits in the transition framework that leads to less success in the transition process. It could be a risk factor in *Agile* transformation process too.

“Even as a proponent of Agile, I have to say, as of today, that besides some Agile projects here and some Scrum projects there, there are not many companies that have been truly able to successfully implement and deploy Agile, Scrum or even Kanban, to their whole IT department’s projects portfolio, much less to their entire company, with clearly recognized and sustainable business benefits.” P2, Agile Coach.

Software companies need to define expected values before starting *Agile* transition and *Agile* transition framework must consider

such values in its workflow. Indeed, the transition framework should be defined in such a way that facilitates achievement of business values as company and teams expect.

5.1.2. Iterative

Most of the participants declared that a successful transition framework should be 'iterative'. They declared that this feature facilitates altering development process.

"It can be a multilevel process, like spiral method, or like a scrum project. Pick up some practices and try to adapt to them, and then start again with next practices. It cannot be done overnight or in short time. Next steps can be based on the successes in previous steps" P21, Scrum Master.

"A scientific, evidence and learning based improvement framework. For example the Shewhart or Deming cycle of Plan-Do-Check-Act." P1, Head of Development Dept.

The idea of PDCA was mentioned by some of the participants too. Furthermore, this approach is close to Agile approach. It means that software companies and organizations can use a simple and well-known idea for transition rather than using a complex and detailed framework; as one of the participants mentioned, "use Agile to go Agile" (P39).

"The most important part is that organization should try to be Agile not do Agile. This makes transition hard, so, may an iterative and continuous approach lead them to a successful Agile adoption." P40, Agile Coach.

As the participants explained, in each iteration only some of the Agile practices may be adopted, evaluated, and adjusted. This is why it seems that this feature can facilitate transitioning to Agile.

"We are talking about people, we cannot change them overnight. Indeed, any change process for people takes a long time. They cannot change at once; change process should be iterative as well as continuous." P17, Project Manager.

Iterative-based framework, as the participants explained, can be useful even for changing people's behaviours and cultures as well as organisational behaviours.

"Many years ago, Toyota with its famous expression, 'Building people before building cars', started Agile movement in industry. They showed that PDCA [Plan, Do, Check, Adjust], leads to a better innovate. PDCA helped them to create a culture of critical thinking and problem solving. Now, this strategy can be used in Agile transition, since Agile is all about change and people." P1, Head of Development Dept.

Moreover, considering this feature in the transition framework can help software companies and teams to remove obstacles they are facing during Agile transition process. This approach is extremely helpful when all parts of a company are not ready to move to Agile.

5.1.3. Continuous

'Continuous' was the third most important characteristic of the Agile transition and adoption framework. Indeed, the participants believed that transition framework should be continuous. This concept is close to iteration. While iteration focuses on repeatable cycle of steps, continuous stressed non-stop cycles.

"... Thinking, investigation, adaptation, evaluation like a chain. There is no specific point that achieving it terminates transition process. You should always adopt yourselves with new changes." P11, Agile Coach.

"I cannot tell what is the best framework for adoption, but generally I think that transition is iterative and should be done in many stages.... continuously and repeatable based on the evaluation of its progress." P3, Project Manager.

One of the participants warned about the lack of this feature in transition framework. Transitioning to Agile should be a continuous process to improve software development process.

"I see companies left and right fail miserably with their Agile transitions due to some assumptions. The second is that the ideal end-state of Agility exists and can be reached. This is more dangerous as the point of Agile is continuous improvement. In general that should be what all organizations are doing. When you combine lack of business goals and continuous structure of your framework, you get a cargo-cult, where an organization pursues the dream of "being Agile" with a belief that once "we're there" by magic all our problems are solved." P1, Head of Development Dept.

Consideration of end-state for Agile transition would be harmful, as one of the participants stated. 'Forever adoption' (P39) may define transition framework better from this perspective.

"Many people ask me about measuring Agile maturity, that's not exactly the right question, because Agile transition is a journey, not a destination. But, I can understand why he asked the question. I should say that transition journey has not an endpoint; it's a forever adoption process." P39, Agile Coach.

Agile transition and adoption framework needs to provide a mechanism in which software companies and teams can adopt their desired and required practices when possible. In this case, transitioning to Agile can be a non-stop process and re-start when a new practice needs to be adopted.

5.1.4. Gradual

The last emerged structural feature of Agile transition and adoption framework was 'gradual'. Data analysis showed that whenever the participants talked about the iterative nature of an effective and applicable Agile transition and adoption, they stressed 'gradual transition' too.

"Organisation should let people change over time. Some behaviours need more time to change. An appropriate action plan can help them to change themselves gradually." P18, Agile Developer.

"Looking at the previous successful transformation shows that those who gradually adopted Agile methods have better performance as well as more positive feeling about their process." P42, Agile Developer.

They stated that gradual adoption is an opportunity for those who are not ready for transition and also is helpful for Agile opponents.

"Most of the companies are not really ready for the move to Agile, especially with all that is required for the transformation at the overall company's or IT's level. Anyway, gradual adoption helps teams to get ready for real transition...." P2, Agile Coach.

Some of the participants explained that gradual transition reduces risks and challenges of the change process, since it gives enough time to people to adapt themselves to the required changes. Indeed, starting transition with simple and easy to adopt practices rather than focusing of more complex practices decreases the probability of failure.

"Go to Agile gradually. I suggest that use Agile methods only for one half of your projects. It is better that you start with some specific and non-critical projects. This strategy helps you to pay attention more to people problems and after success migration, you can involve other parts of your company with less effort and risk. It should be done in more than one step. Plan for some small changes, do them and assess your status. Do this process many times." P4, Project Manager.

Almost all of the participants emphasized that people's mindsets cannot change overnight since it is a gradual process. This approach

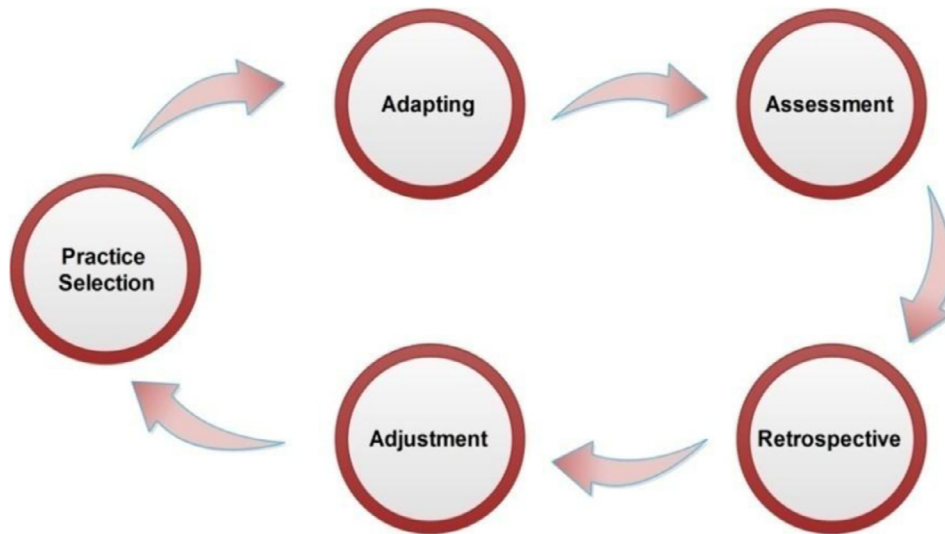


Fig. 5. Agile transition and adoption framework demonstrated by Glaser's temporal/process family.

was consistent with their viewpoints about gradual structure of transition framework too.

5.2. Agile transition and adoption key activities

Data analysis revealed the emergence of several key activities in Agile transition and adoption process. Although, there were some small and detailed tasks, all activities could be classified in a limited number of macro activities as follows, as depicted in Fig. 5.

5.2.1. Practice selection

As explained in Section 5.1, iterative, gradual, and continuous were addressed as some of the main characteristics of Agile transition and adoption framework. Since Agile adoption is better to be gradual, iterative, and continuous, in each round of adoption, only some of the Agile practices would be adapted. The participants recommended picking up only a few practices within a period of adaptation time.

"Pick up some practices and try to adapt to them, and then start again with next practices." P2, Agile Coach.

The participants emphasized that practice selection is so important and critical. Practice selection should be based on the company's real needs and priorities. This feature reflects the value-oriented nature of the transition framework. This is the point where the transition framework focuses on business values.

"We started with Scrum because our main challenge was managing the project and customer collaboration, many other companies have different situations, I mean for example, technical problem or something else." P17, Project Manager.

"At the beginning of our Agile transformation, we only used some practices of Scrum as Agile pattern. It worked well most of the time, but not sometimes. We also realized that our transformation was not iterative and incremental. After attending some workshops, we have decided to focus most on the main points of team. We also added Kanban as an Agile pattern available." P22, Agile Coach.

The participants also declared that, although all Agile methods and practices are useful, each of them needs its own prerequisites and should be used in the right place. Therefore, practice selection is not a simple task and requires particular attention.

"Agile practice and method selection is a big issue even within a company. I am working with a multi-sector company; they use both

Scrum and Kanban. They use them for the reasons. In one part of their company, they cannot do any planning, it is not planable because they do not know the demand yet, for them Kanban is very good, since it is flexible. In [their] product development department of this company, they will use Scrum, because it provides many good activities for customer collaboration. So, even in one company, it is important which methodology we are going to practise. Then in that methodology we should decide that which practices we want to use, e. g. TDD? ... It depends on the many factors." P13, Agile Coach.

"In our company, the developers chose XP because it is most focused on software coding and they had many technical challenges. The product owners, business analysts, etc. became more involved when they introduced Scrum. Infrastructure adopted Kanban." P23, Agile Coach.

Although there are several Agile methods each encompasses multiple practices, not all of them may be suitable for all organizations. Therefore, picking appropriate Agile practices from different methods rather than adapting to a whole Agile method, can be considered as practice selection strategy.

"My strategy is not adopting a whole Agile methods, most often, organizations need only some of the specific Agile practices, regardless of those practices belonging to which Agile methods" P28, Agile Coach.

As the participants stated, some of the key roles like coaches, managers, project managers, or even a particular committee are responsible for practice selection. They should consider several factors for practice selection, such as company's needs and goals, Agile teams' capabilities, customer, project, etc. For instance, if their customer cannot be involved, most of the Scrum practices cannot be adopted well. Therefore, doing an assessment, even a simple assessment, is very helpful. In fact, it can show software companies which Agile practices are suitable for them.

5.2.2. Adaptation

Adoption is defined as the ultimate goal of the Agile transition process. Sustainable adapting strategy will lead to Agile adoption. The participants explained that Agile adaptation cannot be considered as a time-boxed activity. They expressed that Agile adoption and adaptation are nothing more than changing people's mindsets, behaviours, and cultures that is not easy and needs adequate period of time.

"Adoption needs real change. Unfortunately, change is difficult and hard and many people do not have an appetite for it. So, when we are talking about Agile adoption, enough time should be considered. Adapting to new behaviour and mindset is really hard and time consuming." P40, Agile Coach.

"Agile adoption is dealing with human side of change. They resist (against) any changes, unless they feel that they need to change. Pushing people to change is the worst type of adoption, so, adoption takes more time than we expect." P2, Agile Coach.

Some of the participants emphasized institutionalisation of change. They believed that without considering this approach, after some initial success, change efforts eventually fail.

"The only way that an Agile practice can show its advantages is institutionalisation of that practice. We had a good experience about CMMI. For achieving best benefits, Agile practices also need to be institutionalised at organization." P23, Agile Coach.

"Agile practices should be adopted and institutionalised, otherwise, transition will fail." P35, Head of Development Dept.

Some of the participants emphasised that adaptation to new approach is not a team-effort; it should be an organisational effort. They believed that even an 'individual-adoption' can be accepted, but the essence of Agile is 'whole-organisational effort'. Furthermore, some of the participants confessed that in the real environment 'whole-organisational adoption' is so hard and most often some teams remain non-Agile.

"About company, we accepted that some parts of our company remain waterfall. We preferred to preserve our experts rather than push all parts to be Agile at that time." P21, Scrum Master.

"We evolved into Agile, one team at a time. It has to make sense with the workflow and the development cycle so we are not 100% Agile and will probably always have 2+ teams in Waterfall." P33, Senior Manager.

Providing a good coaching and mentoring service during this stage will be so helpful especially where team members are discouraged from the transition challenges and problems and need to be guided and encouraged to adapt to new roles and responsibilities.

5.2.3. Assessment

While the participants were discussing the governance of Agile development, they addressed assessment as an important activity during the transition process. However, the iterative nature of the transition and PDCA approach shows the role of the assessment in ATP.

"Assessment of assets in each company for transformation is necessary. This process needs a plan that asset assessment improves it. Don't fear about change, do it and check your status, if it is not ok yet, change it and do it in other way. Control of transformation process is important and be careful about it." P16, Project Manager.

This activity may help in 'finding the challenges Agile teams are faced with' (P2), 'monitoring the processes and progresses of transition' (P23), and even changing priorities.

"We had a mistake in transformation; I advise others that define a measurement tool for assessing their transformation process. It can help them in challenging barriers and also in achieving their goals." P35, Head of Development Dept.

The participants also addressed several questions which should be answered by assessing the transition progress. These questions may encompass many issues such as development pace, right iteration duration, team size, people collaboration, training, leadership,

team empowerment, challenges and impediments, quality of meetings and retrospective, realising business benefits, and so forth.

"My advice, during transition, is to experiment yourself and see what works for you and where. I would also suggest escaping from the Agile brand and looking towards fields with more permanence. This strategy helps you to find which practices are suitable for you." P6, Agile Consultant.

Although a few of the participants believed that appraisal assessment should be done by an external expert, most of them called this activity a 'self-assessment' process, which can be done by the Agile teams themselves. One of the participants discussed her simple experience-based tool which was proposed for self-assessment along the transition process (P39). Also, two of the participants gave some warnings about this activity: using traditional metrics and paying less attention to business benefits during the assessment process.

"One factor which appears to decelerate Agile adoption is that Agile teams are measured using traditional metrics..." P40, Agile Coach.

"Assessment should follow the business benefits; otherwise, it would be useless and boring." P22, Agile Coach.

Data analysis in this part revealed that there are no common and clear ideas about some questions; Who is responsible for this activity? When should it be performed? and other details associated with assessment. In general, however, most of the participants believed that after adapting some practices or in a specific time period, assessment can be performed. They also mainly addressed coaches or managers as those who are responsible for handling this activity.

5.2.4. Retrospective meeting

The results of assessment which reflect the current state of ATP need to be reviewed and discussed in a retrospective meeting, as the participants mentioned. They also declared that in this meeting, like Scrum, all team members should be involved and practice adaptation should be reviewed.

"After assessment, a retrospective meeting can be held. It should be more interactive session rather than just one person reading down the points from each team. They should brain-storm on the improvements and things which puzzle and note down the action items." P2, Agile Coach.

Group discussion is a key success factor in such a meeting. People's collaboration and involvement make retrospective more effective, since *"the plan for the future should be discussed in this meeting"* (P22).

"Although most of the team members are guessing what works and what does not, retrospective meetings move them from guessing to knowing. It's why I emphasised letting people know the results of assessments. Believe that, the pains of transition cannot be solved by only one member or a group..." P40, Agile Coach.

Furthermore, as the participants stressed, the main goal of this meeting, is finding the challenge areas, weaknesses and strengths of the Agile teams involved in the transition. Moreover, any new idea and suggestion will be welcomed.

"Participants will realise that we are paying attention to the suggestions and things which are being highlighted during this meeting." P13, Agile Coach.

There was no consensus about the time of the retrospective. While many of the participants addressed it after a specific assessment, others believed that it can be based on a pre-defined duration. However, the latter group stressed the necessity of doing assessment before performing the retrospective.

5.2.5. Adjustment

Some of the participants addressed adjustment as another activity of the transition framework. This activity can be considered after retrospective meeting. The main aim of this activity is adjusting and adapting Agile practices to particular conditions.

"After a challenging meeting, we decided to replace product owner with a virtual customer. We had many problems with him and his company. We had not any improvement in customer related practices." P4, Project Manager.

As to a suggestion by some of the participants, if Agile teams have not adapted effectively yet to the selected practices, it is better to try to adjust themselves to the current practices, before picking new ones in the next round.

"I strongly suggest my clients not to pick new practices if they have not adapted previous ones effectively. At first they should adjust themselves to the current practices. Retrospectives show them their weaknesses and give them new opportunity to adapt." P35, Head of Development Dept.

Considering this activity as a distinct activity was a debate. While many of the participants discussed the necessity of adjustment and addressed it as a separate activity without explaining its details, some others believed that adjustment can be done in next iteration.

6. Discussion

In this section, a detailed discussion on comparison between our framework and previous works will be provided. Then, the implications of the results for both theory and practice are discussed. Afterwards, we present how the findings of the study have been evaluated, and finally the limitations and future works of the study are explained.

6.1. Comparison with related works

As explained in Section 3, although there are lots of studies regarding Agile transformation, most of them have focused on some specific parts of the transition and most often have provided only lessons learned and recommendations. However, a few frameworks and models have been proposed for Agile transition and adoption. This section will take a look at them and compare them with the framework emerged in the current study.

Boehm and Turner's framework (Boehm and Turner, 2003) helps software companies to choose either disciplined methods or Agile methods, or a mix of both. Their framework does not explain ATP and so, cannot be considered as a complete Agile transition framework. However, it was the first academic attempt regarding Agile transformation. Compared to that framework, our study focused on transitioning to Agile (not to make a decision about it) and its related aspects including transition framework. Also, our study has tried to explore different aspects of the transition.

Sidky et al. (2007) proposed a stepwise adoption approach that was based on the maximum potential Agility of an organization that suggests some specific Agile practices for adaptation. This framework, as discussed in Section 3, is subjected to serious criticisms about its weaknesses including high complexity, huge amount of required overhead, inconsistency with the nature of Agile approach, the ignorance of considering business values, low flexibility in its application, forcing companies to choose specific practices, and so forth (Esfahani, 2012; Soundararajan and Arthur, 2011; Soundararajan et al., 2012). As mentioned earlier, the main reason for such problems is proposing a strict and highly disciplined framework. The developed framework in our study has none of the above problems; while business values are the core of the transition framework, it gives organisations freedom and flexibility to select their desired practices. Also, it directs

the team to be Agile through an Agile compatible roadmap with the minimum organisational overhead to manage the transition. The proposed framework is applicable in all software companies regardless of size.

Qumer and Henderson-Sellers (2008) proposed Agile Adoption and Improvement Model (AAIM) that defines six levels of Agile adoption, as explained in Section 3. However, their framework, unlike Sidky et al.'s framework does not force companies to choose specific practices for each Agility level, but it is subjected to forcing Agility to specific levels, which leads to less flexibility of the transition and so, it is not compatible with the Agile approach (Soundararajan and Arthur, 2011). This framework also does not explain the procedure of Agile adoption and does not consider the critical factors of the transition and its potential challenges. Advantages of our framework over that framework are the same as mentioned regarding Sidky et al.'s framework.

In another study, Sureshchandra and Shrinivasavadhani (2008) proposed a framework for Agile adoption in distributed development of the structure of which is based on Boehm and Turner's framework. As explained in Section 3, they mostly tried to apply Scrum in distributed companies rather than proposing an Agile adoption framework. However, unlike our framework, they did not address the procedure of adoption and their proposed framework does not consider business values in its nature. Since their proposed framework is the same as Boehm and Turner's framework in its nature, comparing our framework with theirs is the same as explained regarding the Boehm and Turner's framework.

Regarding the three above frameworks, focusing on multi-level adoption is a serious criticism, because such an approach is exactly that of the traditional approach and reminiscent of the role of the CMM and CMMI in software process. Although these models help software companies to improve their development process, they lead to less flexibility and Agility. Our framework does not consider any Agility levels and does not force companies to adapt to particular practices. Conversely, it gives them freedom to choose the practices they think are the best for them.

Using an Adapted Structuration Theory (AST), Cao et al. (2009) showed the important factors that affect the Agile adoption. In a Case Study research involving four companies, they explored the factors that can impress the transition. They also tried to suggest some practices for adoption based on the structure of the Agile methods. Despite the weaknesses of their framework, as explained in Section 3, it was a step forward, since they included human-related factors to Agile transition. However its attempt to force choosing some specific practices without considering the business values required by the company is a big concern in that theory. Also, without providing the guidance on how to do the transition or its procedure is a serious weakness of our framework. Indeed, their study was focused more on how the structure of Agile methods can impress the adaptation rather than proposing an adoption or the transition framework.

The emerged framework of our study illustrates a transition framework which is flexible and easy to use. It also meets the requirement of GT. As Glaser and Strauss state the theory should be "flexible enough to make a wide variety of changing situations understandable...[and] general enough to be applicable to the whole picture" (Glaser and Strauss, 1967). In the next sections, it will be explained that using this framework in practice is flexible and understandable. Furthermore, developing this framework inductively, heralds its usefulness in real environments (Hutchinson, 1993).

6.2. Implications for theory

ATP is a socio-technical process which mostly focuses on people (Highsmith, 2002). Agile manifesto and principles behind it (Beck et al., 2001) present several new values in the software industry. In fact, Agile methods have promised new values to software

companies compared to traditional methods. Although it seems that Agile values and principles are simple, this study supported the previous research studies and showed that a successful transition from traditional to Agile methods is not as easy as people expect and many issues should be considered in this organisational mutation.

This study also showed that a successful transition framework should be iterative, continuous, gradual, and value-based. These characteristics represent the main features of an effective process improvement model and recall the Spiral model too (Boehm, 1988). Each of these characteristics emphasises a critical aspect of Agile transition. ATP is not an overnight project and its progress should be gradual and continuous. Furthermore, since the real aim of the transition is achieving business benefits, the transition framework should be value-oriented. At the same time, the ability to define multiple adaptation iterations assists in prioritising practice adoption and other adoption related tasks.

Emerging stepwise activities of Agile transition and adoption framework was another finding of this study. Although there are no standard activities for Agile adoption, the emerged steps, seem to be sufficiently flexible and applicable to all organizations regardless of their size and projects type. While the main step is Agile adaptation, others play critical roles in Agile transition too. Practice selection by considering constraints and priorities helps to achieve the most Agile values during the adoption iteration cycle. Also, Agile teams need to assess their progress and find their weaknesses and strengths. At the end of every iteration and after doing assessment, a retrospective is held to take a look back at Agile adoption in that iteration and to discuss what was successful and what was unsuccessful about process improvement. Based on the outcomes of retrospective, Agile team may decide to adjust some Agile practices based on their needs and organisational constraints. Adjusted practices need to be adapted in next iterations. Furthermore, providing the transition facilitators in all steps helps involved members to deal with the challenges and learn their new roles. This study also revealed several transition facilitators necessary to support the transition process (Gandomani et al., 2014). The cycle of these activities is completely consistent with the incremental and iterative nature of the Agile approach.

6.3. Implications for practice

This study has substantial implications for practice. The results of this study described a practical Agile transition and adoption framework which is the core of ATP. Software organisations and companies must understand and realise all dimensions of this framework before starting transformation.

6.3.1. Modelling Agile transition and adoption

The results showed that Agile transition can be modelled by PDCA approach (Deming, 2000). This model serves all main characteristics and steps of the emerged framework well. Furthermore, such a framework is consistent with Agile approach; it is enough agile, flexible, visible, and low ceremonies. Fig. 6 illustrates the conceptual model of the emergent framework in the form of PDCA approach including the suggested activities.

Organisations can prepare an action plan based on this approach and utilise it for transformation. Such a framework has minimum overhead and is easy to apply even in small software companies. In this way, each organisation can define a robust fundamental Agile framework.

6.3.2. Proposing a practical Agile transition and adoption

This study proposed a practical framework for Agile transition and adoption on the basis of the substantive data. This framework in a real environment can be applied by Scrum notions. Since Scrum is one of the most popular Agile methods, most of the software engineers

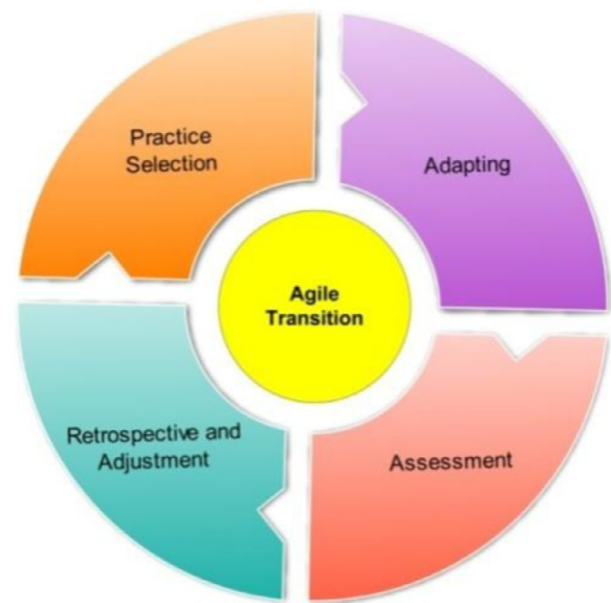


Fig. 6. Agile transition model in form of PDCA approach.

and development teams are familiar with it and its notions. Therefore, they can use Scrum core principles for implementing transition framework. All the Agile practices can be maintained in a 'Practice Backlog'. All practices in 'Practice Backlog' are ordered based on their priorities. Priority of each practice represents the business benefit and value that it can add to the organisation. In each iteration, development teams pick up one or more Agile practices from Practice Backlog and start to adapt to them in a pre-defined time span. At the end of the iteration, teams will be assessed to evaluate their success in practice adaptation. Based on the performed assessment, in a retrospective, they discuss what happened in the iteration and what their suggestions for the next iteration are. In this meeting, they are able to share their experiences in practice adaptation and make better decisions for the next iteration. Also, they may decide on the adjustment of some practices to fit their needs and meet their requirements. Fig. 7 depicts this scenario of Agile transition and adoption framework. ATP needs to be supported by providing transition facilitators as well (Gandomani et al., 2014).

This application of the transition framework preserves all the emerged activities and characteristics too. In fact, Scrum-like framework has been used for Agile transition and adoption process. However, the outcomes of iterations are adopted practices and not deliverable product increments.

6.3.3. Level of Theory

In this study, a middle-range, substantive theory, was developed which covers neither a very narrow, constrained set of phenomena nor a very broad range of phenomena. Such a middle-range theory prevents the "narrow-empiricism" and "theory of everything" criticisms. The proposed framework describes how software companies and teams can change their software development approach and adapt to Agile practices. This research study does not claim the proposed theory to be universally applicable but rather it accurately describes the research phenomena, and it can be further extended when data from new contexts are compared to the existing theory. Therefore, we argue the theory meets Weber's level criterion (Weber, 2012).

7. Evaluating the results

Various perspectives and approaches have been proposed to evaluate a qualitative research study and GT as well

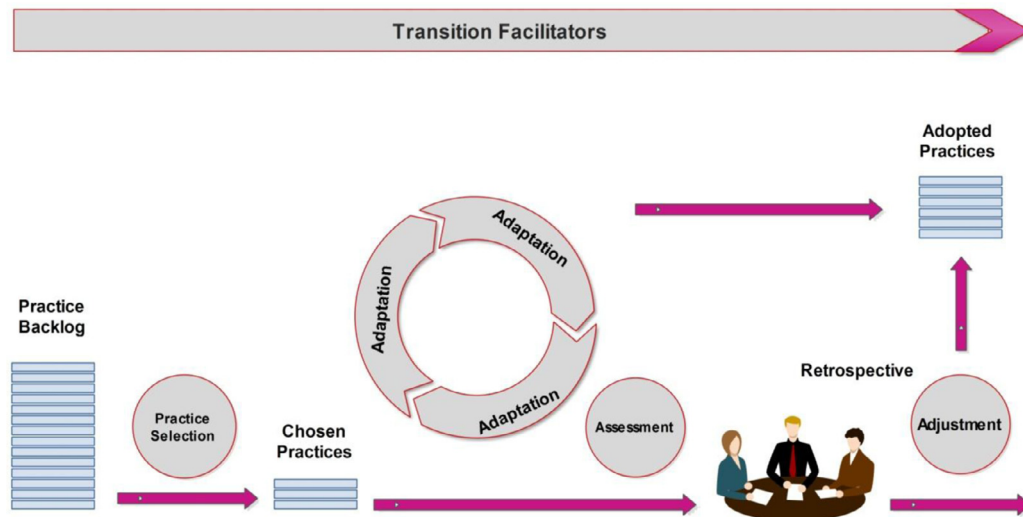


Fig. 7. Practical Agile transition and adoption framework.

(Birks and Mills, 2011; Creswell, 2012). We employed Glaser's approach (Glaser, 1992; Glaser, 1978), Lincoln and Guba's approach (Lincoln and Guba, 1985), Angen's approach (Angen, 2000), and Weber's approach (Weber, 2012) to evaluate the research process and its findings. These made us confident of the credibility and trustworthiness of the findings. Due to space limitation, the rest of this section presents how the findings are evaluated against Glaser's approach and Lincoln and Guba's criteria.

7.1. Glaser's approach

Glaser and Strauss (1967) firmly believe that GT is used to build a theory, not to verify it. Indeed, they believe that validity in its traditional sense is not a critical issue in GT. Therefore, follow-up study(ies) to verify the emergent theory is left to other methods which are better suited to this purpose (Glaser, 1992). Glaser (1992) believes that an emerged grounded theory is grounded in data and in some respects already has been verified. "A grounded theory gets its concepts from the data; it does not bring ideas to force the data that need to be subsequently tested" (Glaser, 1992). Yet, he proposes four different criteria for evaluating credibility of the emergent theory and its related categories: *fit*, *workability*, *relevance*, and *modifiability* (Glaser, 1992; Glaser, 1978).

Fit criterion mainly focuses on emerging codes, categories, and theory from data rather than researchers' preconceived views. It primarily emphasised that GT researchers must set aside their own perceptions and biases and remain open and sensitive to the data (Glaser, 1978; Glaser and Strauss, 1967). Furthermore, emerging the theory with diverse set of data makes it enough close to substantive areas and applicable to dealing with the areas actual realities (Coleman and O'Connor, 2007). Some of the researchers believe that this criterion is the most important ones to evaluate validity of a grounded theory (Lomborg and Kirkevold, 2003).

Workability is used to assess the integration of the core category and its main related categories. On the basis of *workability* a theory should be able to explain what is happening in the area under study and predict what will happen. This criterion tries to show how well the major concerns of the people under study are continually resolved (Glaser, 1992).

Relevance refers to how well a grounded theory's focus on main problems and processes that emerge in a substantive area. Glaser (1978) believes that "Grounded theory arrives at relevance, because it allows core problems and processes to emerge". Indeed, a theory has

good grab for practitioners and is *relevant* when it allows the main problems in the area under study to emerge.

Modifiability focuses on the ability of the theory to be continually modified upon receipt of new data. The rationale behind this is that the emergent theory has only partial closure and adding more data and views can change it. This criterion reflects ever-ongoing nature of GT which has potential for further development (Glaser, 1978). As he states, a category, which fits and works, is relevant and is subject to continual modification and thus, will integrate a theory in the way that the theory is dense and saturated with relationships (Glaser, 1978).

In this research, we avoided carrying out an extensive literature review at the initial stages. Our prior knowledge on ATP was primarily acquired through interaction with the participants while collecting data from them. Also, we employed theoretical sampling—to decide about the next data (what and where) based on the emerging concepts, categories, and theory. Moreover, we particularly tried to recruit participants across different companies in different countries and in different roles primarily to contribute most relevant explanations on the emerging concepts from diverse perspectives. Additionally, we followed Glaser's advice regarding the interviews questions and tried to continually change them in order to solely focus on those concerns which seem central to the emerging concepts, categories, and theory (Glaser, 1978).

Since we continued theoretical sampling until theoretical saturation was noticed in the new data, all the concepts, categories, and their properties that define the grounded theory together, have only emerged from substantive data, without any force by preconceived ideas or hypotheses. Hence, we firmly believe that the emerged theory and its related categories fit the realities under study. GT researchers can assess workability and relevance criteria by asking the participants whether the emerged categories and theory relate to the substantial social problems in the area under study. To do this, while data analysis was on-going and categories were emerging, we attempted to write some technical reports and papers. Then, we provided those participants who were interested, with the written reports and papers and asked them to give feedback on the emerging categories. For example, two of the participants provided their feedback via email on the emerging theory as follows:

"On the surface they seem to map to reality. It would be interesting to see a transformation run using this theory and framework..." P6, Agile Consultant

"In reviewing your findings, I think I've seen or heard about every area of the framework at some point in real life." P1, Head of Development Dept.

Subsequently, we published some of the papers and presented some of them in international symposiums and conferences. Indeed, to date, we have presented various aspects of the emerging theory to the research participants and Agile experts in some online Agile communities and at different international conferences. Fortunately, we have received much feedback from various practitioners that reflect how the emergent theory is useful, meaningful, and applicable to them. The following are some written feedback received from practitioners:

"The characteristics of the transformation framework, as you described, are the main features that Agile coaches need to consider while proposing an action plan for transformation." Practitioner – Australia (in conference)

"It's easy to understand and it is applicable in all environments. I have a good feeling about it. As you mentioned even in small companies such a framework is usable, and to me, it is a great point". Practitioner – New Zealand (in conference)

"I have been struggling with Agile methods for at least nine years. In my opinion, those you have demonstrated are the reality of the Agile transformation in a good abstraction level." Practitioner, Germany (via email)

"To me, your work (the whole theory) is a detailed change model that can work in all companies that intend to change their processes. However, it fits Agile transformation." Practitioner – New Zealand (in conference)

"I participated in two Agile transformations, but I can see there is a lot I can learn from your research" Practitioner – USA (via email)

"I can say that the reality of Agile transition is more complex than we expect. You findings very much support my experience on the whole transition process" Practitioner – USA (via email)

According to the feedback we have received from the various practitioners, we argue that the proposed framework not only works but also is sufficiently relevant to the subject under study in substantive environments. In addition to the above quotations, we received other feedbacks such as *"real-work transition"*, *"easy to learn and follow framework"*, *"rational-based framework"*, etc. made us confident of the emerged framework.

The emergent framework was modifiable during the research study. The theory has evolved over time when we compared new data to the existing concepts and categories. Agile transition and adoption process, its principle characteristics, the transition framework, the transition challenges, facilitators, and prerequisites were evolved and matured through the research process as we went from discussing with junior Agile practitioners to more mature Agile coaches. We strongly believe that the emergent theory and its related categories can continue to evolve by receiving new relevant data and compare them with the existing theory and its related categories, like the developed transition framework. Therefore, the emerged theory and categories are readily modifiable.

7.2. Lincoln and Guba's approach

Lincoln and Guba (Lincoln and Guba, 1985) believe that a qualitative research (like GT) needs to be evaluated by some criteria of trustworthiness including *credibility*, *auditability/dependability*, *transferability*, and *confirmability*. As they confess, *credibility* is the most important criterion among them. Although we adopted all of the above criteria of trustworthiness, the following elucidates how this study has been evaluated against *credibility* criterion.

Lincoln and Guba (Lincoln and Guba, 1985) argue that credibility in establishing trustworthiness can be achieved through several criteria including *triangulation of data*, *peer debriefing*, and *member checking* as follows:

7.2.1. Triangulation of data

We carried out the initial interviews primarily to collect data from the research participants. Then we applied theoretical sampling and interviewed several participants from different roles and responsibilities including project managers, Scrum masters, Agile developers, Agile coaches, etc. in order to obtain a comprehensive view of ATP, the phenomenon under study. Fortunately, attending a variety of Agile practitioners from different countries and continents with different behaviours and cultures provided us a great opportunity to study ATP from different perspectives. Also, since literature is perceived as a great source of data, we conducted a major literature review after sufficiently grounding the emerging categories and concepts to integrate them into the developing theory. Therefore, data triangulation procedure has been used automatically in this study.

7.2.2. Peer debriefing

With the assistance of peer debriefers, we added credibility to this study. We continually discussed GT procedure, its techniques, and emerging categories with other researchers and experts such as Glaser's representatives in order to get feedback to ensure validity and enhance credibility. We provided the peer debriefers with the whole interviews to read it line-by-line, the same as we did to highlight relevant areas. Then, we provided them with the codes and memos and discussed with them about the coding. In case of disagreement on the coding, we requested a third party to judge and approve the coding. The peer debriefers helped us to clarify our interpretation and challenged our assumptions and led to establish credibility of the grounded theory and its related categories.

7.2.3. Member checking

Although most of the interviews were voice recorded, we transcribed them for ease of use in data analysis phase. We adopted several steps to verify the collected data. For example, we provided the participants with their respective interviews and requested and encouraged them to verify the provided data, suggest any required corrections, challenge what we perceived as wrong interpretations, and provide further information. Moreover, we provided them with the prepared manuscripts or published papers in order to get feedback on the emerging concepts, categories, and theory. The received feedbacks were used as another source of data to integrate with the emerging categories. Since all the participants were overseas, member checking was very tedious and time-intensive. Furthermore, a few of the participants did not have an appetite to collaborate in this process and did not reply our request to review their respective interview transcriptions.

8. Limitations

All the emergent codes, concepts and categories and their related properties and subcategories came directly from real data which were collected from Agile experts in real environments, therefore, the results and findings of this study are grounded in the context of the real data (Glaser, 1998). Nonetheless, this article does not claim that its findings are universal, because its access to appropriate resources was limited to those participants who had voluntarily attended to this research. However, it claims that its findings describe and characterise the area under study (Adolph et al., 2008), but the inherent limitation of GT is that the emerged theory is only grounded in the particular contexts investigated in the research (Adolph et al., 2008).

Ascertaining the authenticity of the participants and provided information is an inherent validity risk (James and Bushe, 2006). Moreover, it was difficult to know whether online identity presented to

the researchers was that of the participant or invited alias, or another person using somebody else's online ID (James and Bushe, 2006). In this regard, we tried to reduce this risk using iterative discussions and different questions about the emerging concepts.

9. Future works

This study focused on Agile transition and adoption framework from a general perspective and it could not be so deep in all the emerged related areas. Furthermore, since GT predominantly directs researchers to core category more than other emerged categories, some of the related categories may be left without receiving enough maturity. These categories may have no significant effect on the real aim of this study, but more investigation on them would be helpful. Therefore, several specific studies can be defined as the future works of this study including:

- Application of this framework in several and receiving their feedback;
- Elaborating all the emerged activities in an explanatory research;
- Finding the real impacts of each of the prerequisites on ATP;
- Elaborating the required roles and responsibilities for doing transformation; and
- Tracking business goals during the adoption process.

10. Conclusion

Carrying out a Grounded Theory study with participation of 49 Agile experts from 13 countries, helped us to propose an Agile transition and adoption framework. The study revealed four major characteristics of an Agile transition and adoption framework including value based, interactive, continuous, and gradual.

The proposed framework can be illustrated by the PDCA approach. In this framework, practice selection, adapting, assessment and adjustment should be performed in turn through an iterative cycle. This framework also can be implemented in real environments by applying Scrum framework. Like Scrum, transition can be divided into multiple iterations. In each iteration, some Agile practices will be selected and Agile teams can try to adopt them. Then, after a specific period of time, by performing an assessment, Agile teams go to a retrospective meeting. They can discuss their lessons learned and may adjust the practices to meet their requirements and accommodate constraints. This cycle can be repeated many times for other non-adopted practices.

In sum, the emerged transition framework is flexible for utilisation by all organisations and compared to the previous frameworks is easier to use and has less overhead.

Acknowledgement

We wish to express our special thanks to all the participants. Also, special thanks to Johanna Rothman, Prof. Hossein Saiedian, Dr. Hazura Zulzalil, and others who patiently reviewed the findings and gave excellent feedback. Finally, our most sincere thanks go to the anonymous reviewers for their valuable comments and suggestions to improve the quality of the paper.

References

Abdelnour-Nocera, J., Sharp, H., 2012. Understanding conflicts in agile adoption through technological frames. *Int. J. Sociotechnol. Knowl. Dev.* 4, 29–45.

Adolph, S., Hall, W., Kruchten, P., 2008. A methodological leg to stand on: lessons learned using grounded theory to study software development. In: *Proceeding of 2008 Conference of the Centre for Advanced Studies on Collaborative Research: Meeting of Minds*. ACM, pp. 166–178.

Angen, M.J., 2000. Evaluating interpretive inquiry: reviewing the validity debate and opening the dialogue. *Qual. Health Res.* 10, 378–395.

Beck, K., 2003. *Test Driven Development: By Example*. Addison-Wesley Professional, Boston, MA.

Beck, K., Andres, C., 2004. *Extreme Programming Explained: Embrace Change*, second ed. Addison-Wesley Professional, Boston, MA.

Beck, K., Beedle, M., Bennekum, A.V., Cockburn, A., Cunningham, W., Fowler, M., Grenning, J., Highsmith, J., Hunt, A., Jeffries, R., Kern, J., Marick, B., Martin, R.C., Mallor, S., Shwaber, K., Sutherland, J., 2001. *Agile Manifesto*. www.agilemanifesto.org (Last accessed May 2014).

Birks, M., Mills, J., 2011. *Grounded Theory: A Practical Guide*. SAGE Publications Ltd, Thousand Oaks, California.

Boehm, B., Turner, R., 2003. *Rebalancing your Organization's Agility and Discipline, Extreme Programming and Agile Methods - XP/Agile*. Universe Springer, Berlin, Germany, pp. 1–8.

Boehm, B.W., 1988. Spiral model of software development and enhancement. *Computer* 21, 61–72.

Bringer, J.D., Johnston, L.H., Brackenridge, C.H., 2004. Maximising transparency in a doctoral thesis: the complexities of writing about the use of QSR* NVIVO within a grounded theory study. *Qual. Res.* 4, 247–265.

Bringer, J.D., Johnston, L.H., Brackenridge, C.H., 2006. Using computer assisted qualitative data analysis software to develop a grounded theory project. *Field Methods* 18, 245–266.

Cao, L., Mohan, K., Xu, P., Ramesh, B., 2009. A framework for adapting agile development methodologies. *Eur. J. Inf. Syst.* 18, 332–343.

Chan, F.K.Y., Thong, J.Y.L., 2009. Acceptance of agile methodologies: a critical review and conceptual framework. *Decis. Support Syst.* 46, 803–814.

Cockburn, A., Highsmith, J., 2001. Agile software development: the people factor. *Computer* 34, 131–133.

Cohen, D., Lindvall, M., Costa, P., 2004. An introduction to Agile methods. *Adv. Comput.* 62, 1–66.

Cohn, M., 2009. *Succeeding with Agile: Software Development Using Scrum*. Addison-Wesley Professional, Boston, MA.

Coleman, G., O'Connor, R., 2007. Using grounded theory to understand software process improvement: a study of Irish software product companies. *Inf. Softw. Technol.* 49, 654–667.

Conboy, K., Coyle, S., Wang, X., Pikkarainen, M., 2011. People over process: key challenges in agile development. *IEEE Softw.* 28, 48–57.

Corbin, J.M., Strauss, A.C., 2008. *Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory (3e)*. SAGE Publications Inc., Thousand Oaks, California.

Creswell, J.W., 2012. *Qualitative Inquiry and Research Design: Choosing Among Five Approaches*. Sage Publication, Thousand Oaks, CA.

Creswell, J.W., 2013. *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*, fourth ed. SAGE Publications, Inc., Thousand Oaks, California.

Deming, W.E., 2000. *Out of the Crisis*. The MIT Press, Cambridge, Massachusetts.

di Gregorio, S., 2003. Teaching grounded theory with QSR-NVivo. *Qual. Res. J. Spec. Issue* 2003, 79–95.

Dingsøyr, T., Nerur, S., Balijepally, V., Moe, N.B., 2012. A decade of agile methodologies: towards explaining agile software development. *J. Syst. Softw.* 85, 1213–1221.

Esfahani, H.C., 2012. *Transitioning to Agile: A Framework for Pre-Adoption Analysis using Empirical Knowledge and Strategic Modeling*. Graduate Department of Computer Science, University of Toronto, Canada.

Gandomani, T.J., Zulzalil, H., Abd Ghani, A.A., Sultan, A.B.M., Sharif, K.Y., 2014. Exploring facilitators of transition and adoption to agile methods: a grounded theory study. *J. Softw.* 7, 1666–1678.

Gandomani, T.J., Zulzalil, H., Ghani, A.A.A., Sultan, A.B.M., Parizi, R.M., 2015. The impact of inadequate and dysfunctional training on Agile transformation process: a Grounded Theory study. *Inf. Softw. Technol.* 57, 295–309.

Gandomani, T.J., Zulzalil, H., Ghani, A.A.A., Sultan, A.M., Nafchi, M.Z., 2013. Obstacles to moving to agile software development: at a glance. *J. Comput. Sci.* 9, 620–625.

Ganesh, N., Thangasamy, S., 2012. Lessons learned in transforming from traditional to agile development. *J. Comput. Sci.* 8, 389–392.

Ghanam, Y., Maurer, F., Abrahamsson, P., 2012. Making the leap to a software platform strategy: issues and challenges. *Inf. Softw. Technol.* 54, 968–984.

Giles, T., King, L., de Lacey, S., 2013. The timing of the literature review in grounded theory research: an open mind versus an empty head. *Adv. Nurs. Sci.* 36, E29–E40.

Glaser, B., 1992. *Basics of Grounded Theory Analysis: Emergence vs. Forcing*. Sociology Press, Mill Valley, CA.

Glaser, B., 1998. *Doing Grounded Theory: Issues and Discussions*. Sociology Press, Mill Valley, CA.

Glaser, B., Strauss, A., 1967. *The Discovery of Grounded Theory: Strategies for Qualitative Research*. Aldine Transaction, Chicago.

Glaser, B.G., 1978. *Theoretical Sensitivity: Advances in the Methodology of Grounded Theory*. The Sociology Press, Mill Valley, CA.

Glaser, B.G., 2005. *The Grounded Theory Perspective III: Theoretical Coding*. Sociology Press, Mill Valley, CA.

Hajjidiab, H., Taleb, A.S., 2011. Adopting agile software development: issues and challenges. *Int. J. Manag. Value Supply Chains (IJMVSC)* 2, 1–10.

Hammersley, M., Atkinson, P., 2007. *Ethnography: Principles in Practice*, third ed. Routledge, London.

Highsmith, J.A., 2002. *Agile Software Development Ecosystems*. Addison-Wesley Professional, Boston, MA.

Hoda, R., Noble, J., Marshall, S., 2010. Using grounded theory to study the human aspects of software engineering. In: *Proceeding of Human Aspects of Software Engineering*. ACM, pp. 1–2.

Hoda, R., Noble, J., Marshall, S., 2011a. Developing a grounded theory to explain the practices of self-organizing Agile teams. *Emp. Softw. Eng.* 17, 609–639.

Hoda, R., Noble, J., Marshall, S., 2011b. The impact of inadequate customer collaboration on self-organizing Agile teams. *Inf. Softw. Technol.* 53, 521–534.

- Hutchinson, S.A., 1993. Grounded theory: the method. In: Munhall, P.L., Boyd, C.O. (Eds.), *Nursing Research: A Qualitative Perspective*. National League for Nursing, New York, pp. 1212–1810.
- Hutchison, A.J., Johnston, L.H., Breckon, J.D., 2010. Using QSR-NVivo to facilitate the development of a grounded theory project: an account of a worked example. *Int. J. Soc. Res. Methodol.* 13, 283–302.
- Iivari, J., Iivari, N., 2011. The relationship between organizational culture and the deployment of agile methods. *Inf. Softw. Technol.* 53, 509–520.
- James, N., Bushe, H., 2006. Credibility, authenticity and voice: dilemmas in online interviewing. *Qual. Res.* 6, 403–420.
- Jones, M.R., Karsten, H., 2008. Giddens's structuration theory and information systems research. *MIS Q.* 32, 127–157.
- Kelle, U., 1995. *Computer-Aided Qualitative Data Analysis: Theory, Methods and Practice*. SAGE Publications Ltd, London.
- Latino, R.J., Latino, K.C., Latino, M.A., 2013. *Root Cause Analysis: Improving Performance For Bottom-Line Results*. CRC press.
- Lincoln, Y.S., Guba, E.G., 1985. *Naturalistic Inquiry*. SAGE Publications, Inc., Beverly Hills, CA.
- Lomborg, K., Kirkevold, M., 2003. Truth and validity in grounded theory—a reconsidered realist interpretation of the criteria: fit, work, relevance and modifiability. *Nursing Philos.* 4, 189–200.
- Martin, A., Biddle, R., Noble, J., 2009. The XP customer team: A Grounded Theory. In: *Proceeding of Proceedings of the 2009 Agile Conference*, pp. 57–64.
- Mišić, V.B., 2005. Perceptions of extreme programming: a pilot study. In: *Proceeding of IEEE International Engineering Management Conference (IEMC 2005)*, pp. 307–312.
- Misra, S.C., Kumar, V., Kumar, U., 2009. Identifying some important success factors in adopting agile software development practices. *J. Syst. Softw.* 82, 1869–1890.
- Nerur, S., Mahapatra, R., Mangalaraj, G., 2005. Challenges of migrating to agile methodologies. *Commun. ACM* 48, 72–78.
- Parry, K.W., 1998. Grounded theory and social process: a new direction for leadership research. *Leadership Quart.* 9, 85–105.
- Pikkarainen, M., Salo, O., Kuusela, R., Abrahamsson, P., 2012. Strengths and barriers behind the successful agile deployment—insights from the three software intensive companies in Finland. *Emp. Softw. Eng.* 17, 675–702.
- Poppendieck, M., Poppendieck, T., 2009. *Leading Lean Software Development: Results Are not the Point*. Addison-Wesley Professional, Boston, MA.
- Qumer, A., Henderson-Sellers, B., 2008. A framework to support the evaluation, adoption and improvement of agile methods in practice. *J. Syst. Softw.* 81, 1899–1919.
- Qumer, A., Henderson-Sellers, B., McBride, T., 2007. Agile adoption and improvement model. In: *Proceeding of 4th European and Mediterranean Conference on Information Systems, EMCIS 2007*, pp. 21–29.
- Rohunen, A., Rodriguez, P., Kuvaja, P., Krzanik, L., Markkula, J., 2010. Approaches to agile adoption in large settings: a comparison of the results from a literature analysis and an industrial inventory. In: *Proceeding of 11th international conference on Product-Focused Software Process Improvement*. Springer-Verlag, pp. 77–91.
- Salo, O., Abrahamsson, P., 2007. An iterative improvement process for agile software development. *Softw. Process Improv. Prac.* 12, 81–100.
- Schova, Z., 2010. Hidden gap in agile methods adoption. In: *Proceeding of 9th International Conference on New Trends in Software Methodologies, Tools and Techniques, SoMeT_10*. IOS Press, pp. 346–356.
- Sidky, A., 2007. *A Structured Approach to Adopting Agile Practices: The Agile Adoption Framework*. Faculty of Engineering, Virginia Polytechnic Institute and State University, Virginia, USA.
- Sidky, A., Arthur, J., Bohner, S., 2007. A disciplined approach to adopting agile practices: the agile adoption framework. *Innovations Syst. Softw. Eng.* 3, 203–216.
- Soundararajan, S., Arthur, J.D., 2011. A structured framework for assessing the “goodness” of agile methods. In: *Proceeding of 18th IEEE International Conference and Workshops on Engineering of Computer-Based Systems, ECBS 2011*, pp. 14–23.
- Soundararajan, S., Arthur, J.D., Balci, O., 2012. A methodology for assessing agile software development methods. In: *Proceeding of Agile Conference, Agile 2012*, pp. 51–54.
- Srinivasan, J., Lundqvist, K., 2010. Agile in India: Challenges and lessons learned. In: *Proceeding of 3rd India Software Engineering Conference, ISEC'10*. ACM, pp. 125–130.
- Stern, P.N., 1994. Eroding Grounded Theory. In: Morse, J.M. (Ed.), *Critical Issues in Qualitative Research Methods*. SAGE Publications, Inc., Thousand Oaks, CA, pp. 210–223.
- Strauss, A.C., Corbin, J.M., 1990. *Basics of Qualitative Research: Grounded Theory Procedures and Techniques*, second ed. SAGE Publications, Inc., Thousand Oaks, California.
- Sureshchandra, K., Shrinivasavadhani, J., 2008. Adopting agile in distributed development. In: *Proceeding of 3rd IEEE International Conference Global Software Engineering, ICGSE 2008*, IEEE Computer Society, pp. 217–221.
- Tolfo, C., Wazlawick, R.S., Ferreira, M.G.G., Forcellini, F.A., 2011. Agile methods and organizational culture: reflections about cultural levels. *J. Softw. Main. Evol.* 23, 423–441.
- VersionOne, 2013. 7th annual state of agile software development <http://www.versionone.com/state-of-agile-survey-results/> (Last accessed March 2013).
- VersionOne, 2014. 8th annual state of agile <http://stateofagile.versionone.com/> (Last accessed May 2014).
- Vijayarathy, L., Turk, D., 2012. Drivers of agile software development use: dialectic interplay between benefits and hindrances. *Inf. Softw. Technol.* 54, 137–148.
- Wang, X., Conboy, K., Pikkarainen, M., 2012. Assimilation of agile practices in use. *Inf. Syst. J.* 22, 435–455.
- Weber, R., 2012. Evaluating and developing theories in the information systems discipline. *J. Assoc. Inf. Syst.* 13.
- West, D., Grant, T., Gerush, M., D'silva, D., 2010. Agile development: mainstream adoption has changed agility. *Forrester Res.* 2, 41.
- Woodman, M., Moteleb, A.A., 2009. Grounding and making sense of agile software development. In: *Proceeding of 11th International Conference on Enterprise Information Systems-ICEIS 2009*, pp. 234–240.
- Yin, R.K., 2013. *Case Study Research: Design and Methods*, fourth ed. SAGE Publications, Inc., Thousand Oaks, California.

Taghi Javdani Gandomani received his PhD in Software Engineering from University of Putra Malaysia, Malaysia. His research interests in software engineering are Agile software development, development methodologies and empirical studies. He is an Assistant Professor in Islamic Azad University, Iran and has more than 15 years industry experience in software development.

Mina Ziaei Nafchi received the MS in Software Engineering from University of Putra Malaysia. Currently, she is a PhD candidate in Software Engineering at University of Putra Malaysia. Her research area is Agile Software Development.